UNIVERSITY OF PADOVA
Department of Mathematics
Master Thesis in Data Science

Final Project

Overview of Forecasting Methods for Electric and Hybrid Vehicle Licenses in the UK

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1 Introduction

In recent years, the adoption of electric and hybrid vehicles in the United Kingdom has accelerated, supported by environmental policies. Forecasting the number of vehicle licenses for these categories is important for policymakers and the automotive sector, enabling strategic planning and evaluation of green initiatives. This study aims to explore different methods applied to quarterly data on the number of licenses for electric and hybrid vehicles in the UK, covering the period from Q1 2016 to Q2 2024. The analysis was conducted separately for electric and hybrid vehicles. Our goal includes understanding the trend dynamics and exploiting different models to forecast the number of licenses. The dataset used is provided by the UK Department for Transport and contains quarterly registrations of electric and hybrid cars. We evaluate forecasting performance using criteria such as the Akaike Information Criterion (AIC) and Mean Percentage Error(MPE).

1.1 Data Exploration

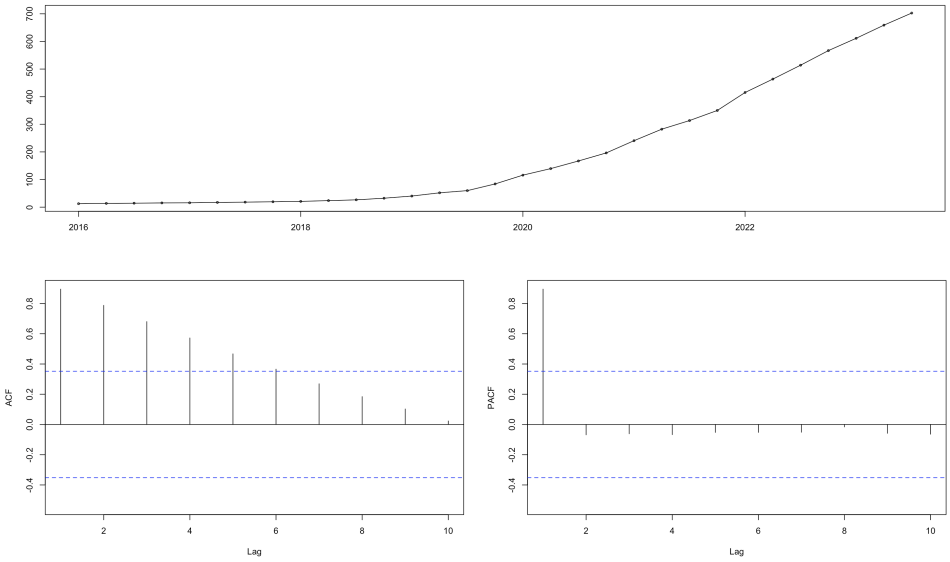

Figure 1: Electric car number of licences (in thousands) and corresponding ACF and PACF plots from Q1-2016 to Q2-2024,UK.

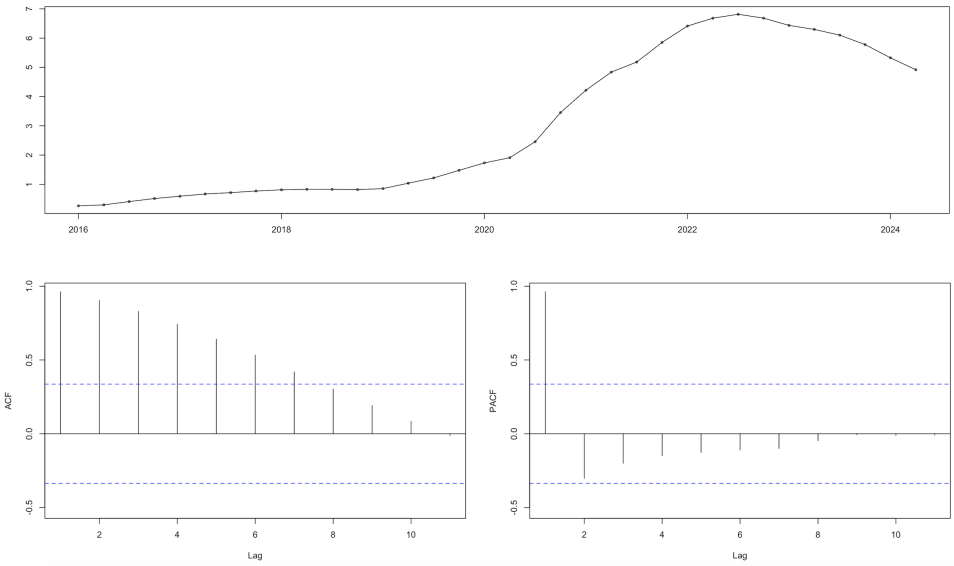

Figure 2: Hybrid car number of licences (in thousands) and corresponding ACF and PACF plots from Q1-2016 to Q2-2024,UK.

Figures 1 and 2 show the quarterly time series of electric and hybrid vehicle licenses in the UK from Q1 2016 to Q2 2024, expressed in thousands of units. Electric vehicle licenses exhibit persistent exponential growth from 2020 onward, while hybrid ones show a strong increase until 2022, followed by a decline. These dynamics justify separate modeling approaches for each series. ACF and PACF plots reveal strong autocorrelation in the early lags of both series, indicating the presence of temporal dependence. Appropriate differencing or smoothing was needed to achieve stationarity. To stabilize the variance and facilitate modeling, both series were log-transformed. The structure of ACF and PACF guided the identification of ARIMA model orders and the evaluation of residuals in the models considered.

Explanatory Variables

In addition to the main response variables, several covariates were considered as potential explanatory variables for the regressors-based models. These include:

- **elec_consumption**: Total electricity consumption in the UK (in TWh),
- **charges**: Number of public charging points
- **uk_petrol**: Average UK petrol prices (in pence per litre)
- **cpi_energy**: Percentage change in the Consumer Price Index for energy
- **cpi_new_cars**: Change in the CPI for new cars
- **brent_oil**: Brent crude oil prices (USD)
- **boe_quarterly**: Bank of England base rate (percent)
- **unemployment_uk**: UK unemployment rate (percent)
- **loans_quarterly**: Average interest rate on consumer loans
- **plug_in_vehicles**: Quarterly number of registered plug-in vehicles

These variables were selected based on economic plausibility, data availability, and their potential contribution about the adoption of electric and hybrid vehicles. They were preprocessed and aligned with the quarterly frequency of the response variables.

2 Methods

2.1 ARIMA-based Models

Autoregressive Integrated Moving Average models are widely used for time series forecasting in the presence of non-stationarity. In our study, we applied ARIMA-based models to both electric and hybrid vehicle license series. Stationarity was achieved through log-transformation and first differencing. For the electric vehicle series, the seasonal pattern justified the use of a SARIMA model, while for the hybrid series, a standard ARIMA was sufficient. The final selection of the model was based on the AIC and diagnostic check of residuals.

2.2 Exponential Smoothing

Among the exponential smoothing methods tested for short-time forecasting, the damped Holt model(with log transformation and forecast back-transformed to the original scale) exhibited the best performance for both electric and hybrid license series. This model outperformed both the standard Holt and Holt-Winters approaches. Additionally,residuals diagnostics showed no significant autocorrelation according to the Ljung-Box test, confirming the model's adequacy with respect to a short-term horizon. The damped version addresses the limitation of the standard Holt method, which assumes an unrealistic, indefinitely increasing, or decreasing trend.

2.3 GAM

The Generalized Additive Model(GAM) is a model that extends the classic linear regression:

$$y_t = \beta_0 + \beta_1 X_{1t} + \ldots + \beta_n X_{nt} + \varepsilon_t$$

and evolves into:

$$y = \beta_0 + \beta_1 f(x_1) + \beta_2 f(x_2) + \cdots + \beta_n f(x_n) + \varepsilon$$

Lastly, the model incorporates nonlinear transformations of the covariates through smooth functions, such as local regression, regression splines, or smoothing splines. The additive structure of the model is particularly appealing as it enables the interpretation of the marginal effect of each explanatory variable. This formulation allows for a high degree of flexibility in capturing complex non-linear relationships while maintaining interpretability.

The adopted approach involves splitting the data set into training and test sets for both hybrid and electric vehicle series. The model was then estimated on the training set. To address autocorrelation in the residuals and account for temporal dependencies, ARIMA models were fitted to the residuals. The resulting forecasts were obtained by combining the Generalized Additive Model (GAM) with the ARIMA components. Finally, the predictions were transformed back to the original scale.

2.4 ARIMAX

The ARIMAX(AutoRegressive Integrated Moving Average with Exogenous Variables) model is an extension of the standard ARIMA framework that incorporates external regressors to improve forecasting accuracy. This model is well-suited for forecasting time series that depend on both their past values and external factors. In our study, a rolling window with an expansive approach was adopted to simulate a real-time forecasting scenario. The rolling window is a technique which is commonly used in time series analysis to assess model performance over time. The idea is to define a fixed or expanding moving training window that progresses through the data: the model is trained on the current active window, a forecast is generated for the next time point, i.e. $t + 1$. Then the window is shifted forward to include the new observation. The process is repeated iteratively and represents a reasonable choice to allow the model to capture progressively the exploding trend. Prior to model estimation, we addressed multicollinearity among explanatory variables by exploiting the Variance Inflation Factor(VIF). Variables presenting VIF greater than 5 were excluded from the model. Subsequently, a backward stepwise selection was applied to eliminate non-significant regressors, ensuring model parsimony. As in previous modeling frameworks, ARIMA models were also fitted to the residuals in order to account for remaining temporal dependencies, and the final forecast was transformed back to the original scale.

3 Numerical Experiments and Results

In this section, we report the numerical experiments and relative results regarding the performed analysis. We divide the presentation of the results for electric and hybrid time series, and then we will provide a comparison.

3.1 Electric

3.1.1 Arima-based

The selected model for the electric vehicle license series is a SARIMA(1,1,1)(1,0,0)$_{[4]}$, estimated on the log-transformed data.

The estimated equation of the model is the following:

$$(1 - \phi_1 B)(1 - \Phi_1 B^4)(1 - B)\log(y_t) = (1 + \theta_1 B)\varepsilon_t$$

where $\phi_1 = 0.943$, $\Phi_1 = 0.296$, and $\theta_1 = -0.402$.

Although the non-seasonal AR and MA terms are statistically significant, the seasonal AR(1) component (Φ_1) has a relatively large standard error (0.177), indicating low significance. Therefore, the inclusion of the seasonal component should be interpreted with caution, and the model behaves similarly to a non-seasonal ARIMA(1,1,1) in practice.

The residuals passed standard diagnostics, as shown in Figure 3, and the Ljung-Box test confirmed the absence of significant autocorrelation ($p = 0.453$).

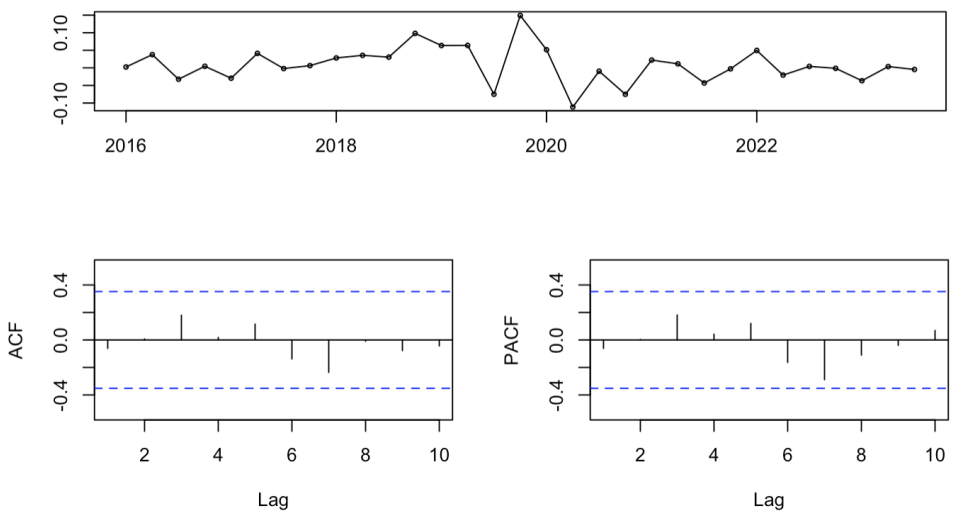

Figure 3: Residual diagnostics for ARIMA(1,1,1)(1,0,0)[4] model (electric vehicles)

Despite their forecasting ability, Arima-based models are purely univariate, meaning that they do not account for potential external factors that could influence vehicle adoption.

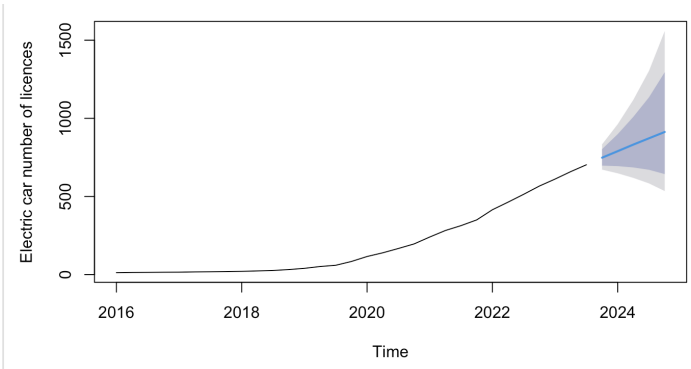

Figure 4: Forecast from ARIMA model for electric vehicle licenses (original scale,thousands of units)

The forecast in Figure 4 shows a continuation of the increasing trend observed in the historical data. The back-transformation from the log scale to the original scale was performed using the exponential function. However, due to Jensen's inequality, we have $\mathbb{E}[\exp(\hat{y})] \geq \exp(\mathbb{E}[\hat{y}])$, meaning that the forecast on the original scale is biased upward and exhibits increased variance. This effect becomes more pronounced when the residual variance on the log scale is not negligible and should be taken into account when interpreting prediction intervals.

3.1.2 Exponential smoothing

Among the exponential smoothing methods tested for the electric vehicle license series, the damped Holt method applied to log-ts provided the best forecast accuracy. The estimated model captures the local trend effectively while mitigating the unrealistic long-term infinite growth of Standard Holt's method. The estimated equation of the model is:

$$\log(\hat{y}_{t+h}) = \ell_t + \phi^h b_t$$

where $\ell_t = 2.437$, $b_t = 0.100$, and $\phi = 0.913$. The forecasts were then exponentiated to return to the original scale.

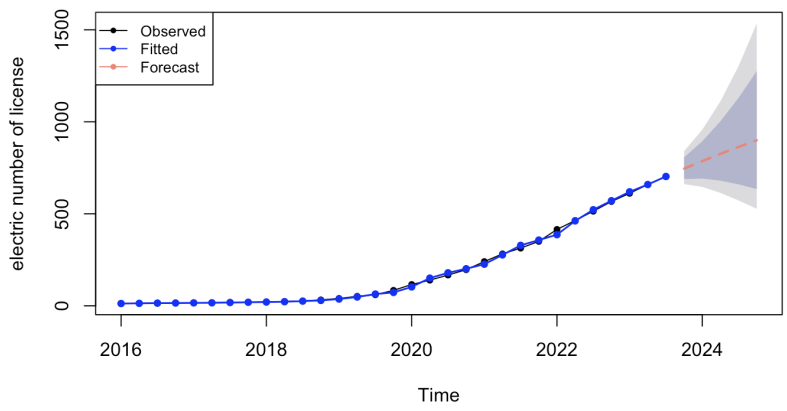

Figure 5: Forecast from damped Holt's method for electric vehicle licenses in thousands (original scale)

As shown in Figure 5, the forecast continues the upwards trajectory while allowing for increased uncertainty in the long term, as reflected by the widening confidence intervals. As in the Arima-based analysis, the results' variance is slightly inflated due to the non-linear transformation.

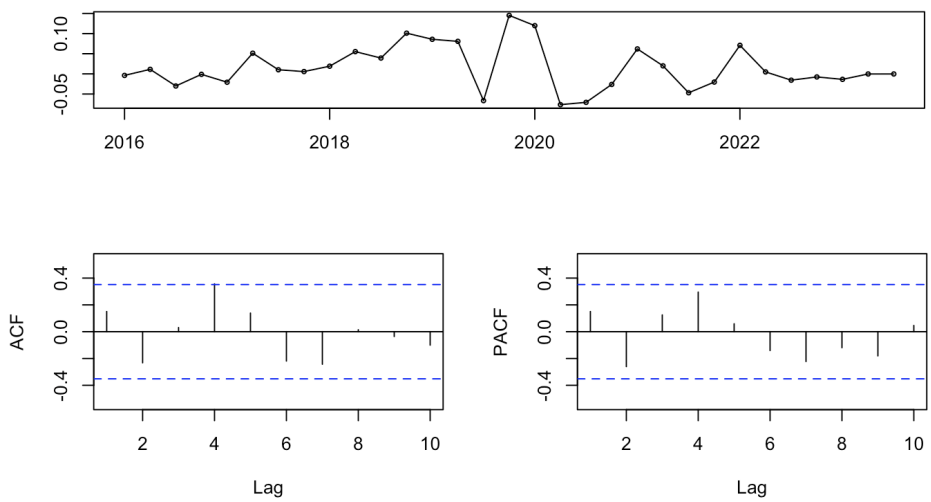

Figure 6: Residual diagnostics for damped Holt's method (electric series)

Figure 6 confirms that the residuals of the model do not show evidence of autocorrelation. The Ljung-Box test yields a p-value of 0.115, supporting the WN distribution of the residuals.

Table 1: Forecast accuracy metrics for exponential smoothing models – Electric series (after scaling back)

Model	RMSE	MAE	MAPE	ACF1	Ljung-Box p	Theil's U
Holt Damped	8310	5370	4.05%	0.05	0.1148	0.40
Holt	9720	7020	4.20%	0.15	0.508	0.39
Holt-Winters	16171	9522	5.03%	0.30	0.109	0.44

The above table supports the choice of the damped Holt model,which showed the best overall performance in terms of RMSE, MAE, and MAPE among the exponential smoothing methods tested.

3.1.3 GAM

The analysis began by applying a log-transformation to the response variable in order to stabilize the variance. The theoretical formulation of the estimated model is as follows:

$$\log(\text{electric}) = \beta_0 + f_1(\text{time}) + f_2(\text{plug_in_vehicles}) + \varepsilon$$

The fitted model was specified as:

$$\log(\text{electric}) = \beta_0 + s(\text{time}, k = 10) + s(\text{plug_in_vehicles}, k = 6) + \varepsilon$$

The smooth term for `time` is highly significant ($p < 2\times10^{-16}$), suggesting a strong nonlinear effect. The smooth term for `plug_in_vehicles` is not statistically significant ($p = 0.117$). However, its exclusion led to a substantial deterioration in model fit, as indicated by standard performance metrics; therefore, it was retained in the final specification. Diagnostic analysis revealed that the residuals exhibited significant autocorrelation, as confirmed by the Ljung-box test ($p = 0.03497$). To address this issue, we modeled the residuals using an ARIMA(2,0,1) process, selected through grid search on AIC values for the ARIMA-SARIMA applied to the residuals combined with ACF/PACF behaviors study. After this correction, the Ljung-Box test on the final residuals gave a p-value of 0.1201, indicating no remaining autocorrelation.

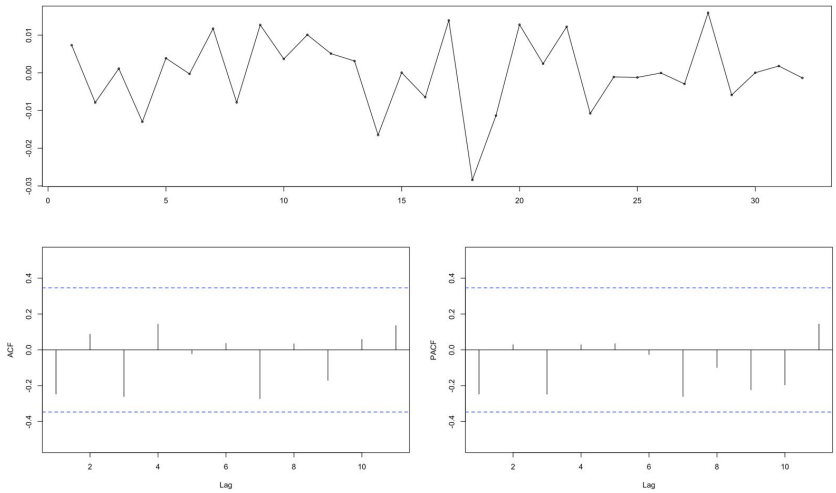

Figure 7: Residual diagnostics for ARIMA(2,0,1) model fitted on GAM (electric vehicles)

The final residuals appear to be white noise, confirming the adequacy of the ARIMA correction.

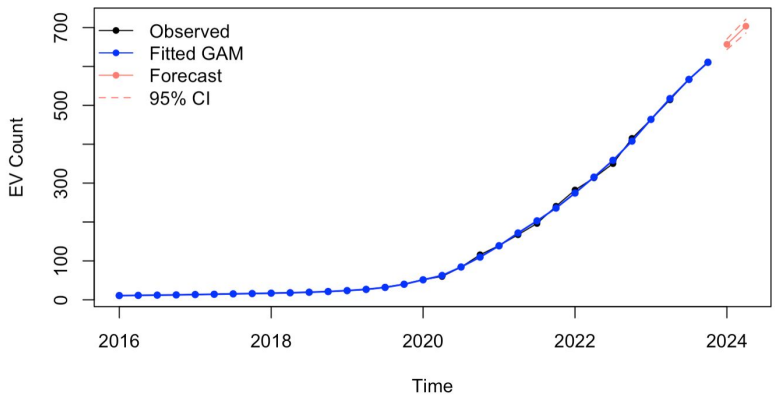

Figure 8: Forecast from GAM+ARIMA (residuals) method for electric vehicle licenses in thousands (original scale)

The plot shows the number of electric vehicle licenses predicted using the GAM + ARIMA approach, which combines nonlinear trends and residual dynamics. The prediction is presented in the original scale, offering a realistic projection based on past data.

3.1.4 ARIMAX

The modeling process began with the identification of relevant and available explanatory variables potentially useful for forecasting. To address multicollinearity, we implemented a function to compute the Variance Inflation Factor (VIF), removing variables with a VIF greater than 5.

Subsequently, a backward stepwise selection procedure was applied to the remaining variables in order to eliminate non-significant predictors, ensuring model parsimony and statistical relevance. The theoretical form of the regressor-based model is given by:

$$y_t = \beta_0 + \beta_1 \cdot \text{elec_consumption}_t + \beta_2 \cdot \text{charges}_t + \beta_3 \cdot \text{uk_petrol}_t + \beta_4 \cdot \text{cpi_energy}_t + \beta_5 \cdot \text{unemployment_uk}_t + \varepsilon_t$$

The estimated model is the following.

$$\begin{aligned}\hat{y}_t &= 3.832 - 0.112 \cdot \text{elec_consumption}_t + 0.092 \cdot \text{charges}_t \\ &\quad - 0.00082 \cdot \text{uk_petrol}_t + 0.024 \cdot \text{cpi_energy}_t + 0.523 \cdot \text{unemployment_uk}_t + \hat{\varepsilon}_t\end{aligned}$$

All estimated coefficients are statistically significant at conventional levels. In particular, `charges` ($p < 2 \times 10^{-16}$) and `cpi_energy` ($p = 3.66 \times 10^{-9}$) emerge as strong positive predictors, suggesting that infrastructure availability and rising energy prices support electric vehicle adoption.

`elec_consumption` shows a significant negative effect ($\hat{\beta} = -0.1121$), while `uk_petrol` has a small but significant negative impact ($\hat{\beta} = -0.00082$), indicating a modest substitution effect with traditional fuels.

Interestingly, `unemployment_uk` is positively associated with the log-response variable, which may reflect indirect macroeconomic influences or policy interventions affecting the market. The residual analysis revealed that the residual variance was not constant, indicating the presence of heteroscedasticity. If the primary objective were inference using multiple linear regression, we would consider applying more accurate transformations on the response. However, since our objective is forecasting, we proceeded by fitting an ARIMAX model to capture temporal dependencies and improve predictive accuracy.

To further improve forecast accuracy, we implemented a rolling window approach, which yielded the best forecasting performance. We used a grid search based on AIC values and observed the behavior of the ACF/PACF plots to identify the optimal ARIMAX configuration. The selected model was: SARIMAX(2,1,0)(1,1,0)[4].

The residuals from this model showed no significant autocorrelation or heteroscedasticity. The Ljung–Box test returned a p-value of 0.5782, confirming that the residuals resemble white noise and behave well.

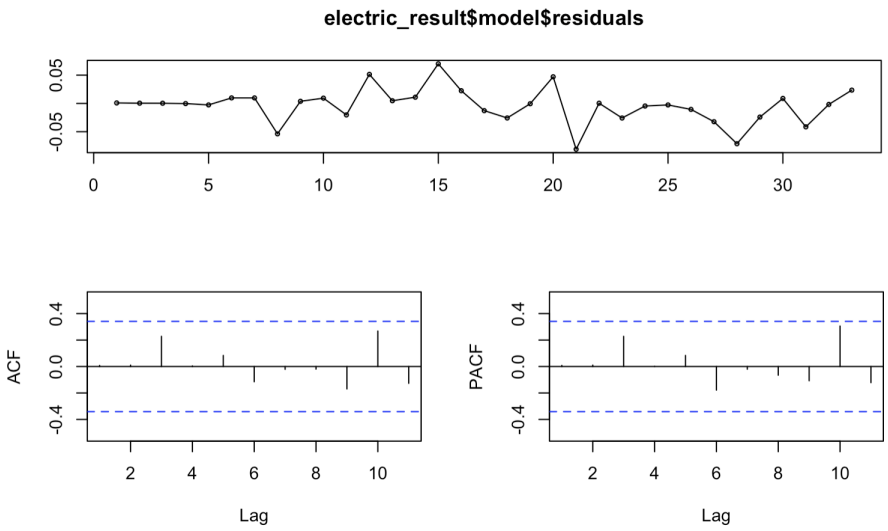

Figure 9: Residual diagnostics for ARIMAX(2,1,0)(1,1,0)[4] model (electric vehicles)

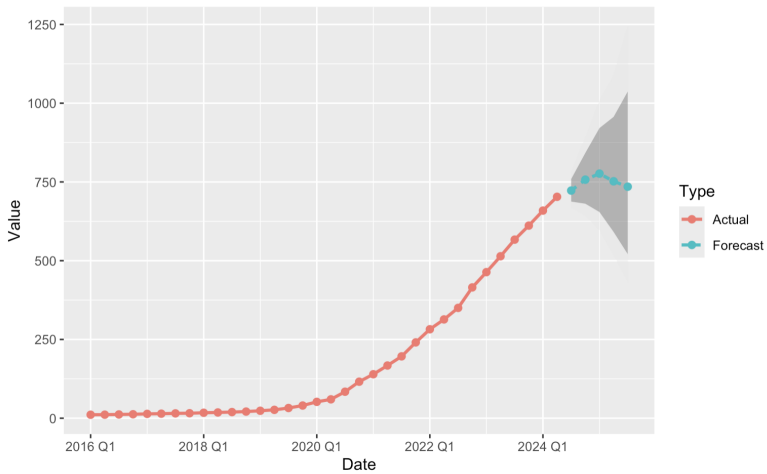

Figure 10: Forecast from ARIMAX model for electric vehicle licenses in thousands (original scale)

The graph displays the observed values (in red) and the forecasted values (in blue). The shaded area represents the 95% confidence interval, which reflects the increased variance resulting from the back-transformation of the forecasts to the original scale. However, the model revealed good metrics with respect to forecasting accuracy on its original scale.

3.2 Hybrid

3.2.1 Arima-based

The selected model for the hybrid vehicle license series was an ARIMA(1,1,1), estimated in the log-transformed data.

The estimated model is the following.

$$(1 - \phi_1 B)(1 - B)\log(y_t) = (1 + \theta_1 B)\varepsilon_t$$

with $\phi_1 = 0.767$ (significant) and $\theta_1 = 0.318$ (not significant). Given the lack of statistical significance of the MA(1) term, the model behaves effectively as a simpler ARIMA(1,1,0). The residual analysis supports the adequacy of the model, as shown in Figure 11. The ACF and PACF plots reveal no relevant autocorrelation and the Ljung-Box test p-value confirms that the residuals resemble white noise ($p = 0.321$).

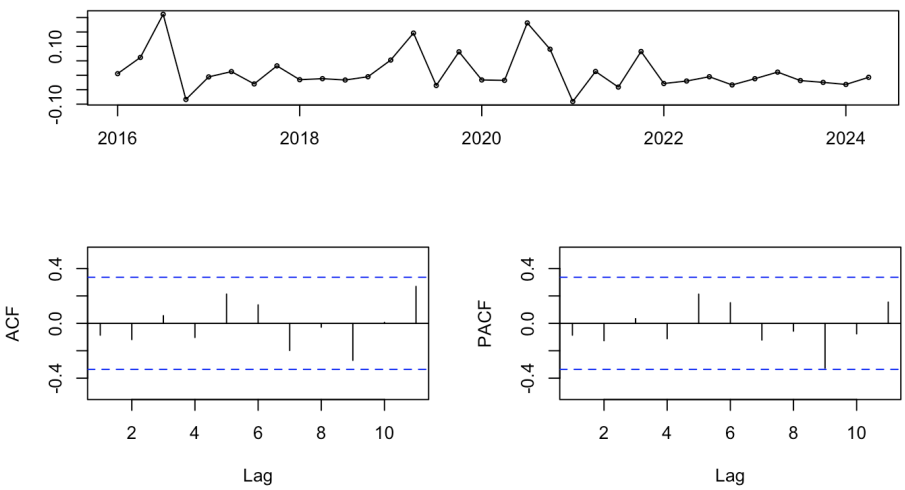

Figure 11: Residual diagnostics for ARIMA(1,1,1) model (hybrid vehicles)

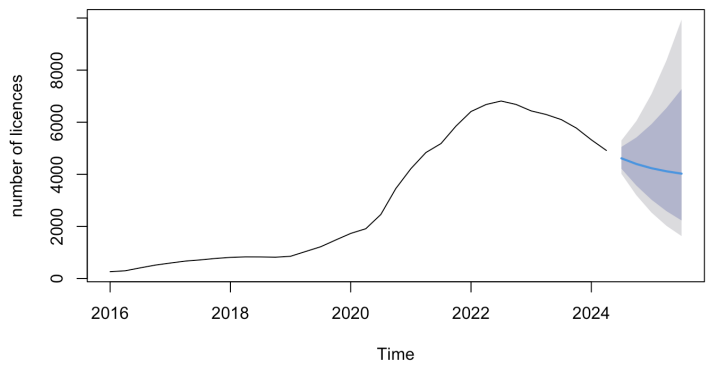

Figure 12: Forecast from ARIMA model for hybrid vehicle licenses (original scale)

The forecast in Figure 12 projects a slight continuation of the recent downward trend. As in the electric case, the forecast has been back-transformed from the log-scale to the original, which tends to inflate variance due to the nonlinear transformation.

3.2.2 Exponential smoothing

Among the exponential smoothing methods tested for the hybrid vehicle ts, the damped Holt method applied to the logarithmic transformation of the data showed the best overall performance, as for the analysis of electric car licenses. The estimated forecasting equation is the following.

$$\log(\hat{y}_{t+h}) = \ell_t + \phi^h b_t$$

where $\ell_t = 5.411$, $b_t = 0.226$, and $\phi = 0.918$. The final forecasts were then exponentiated as done for the electric.

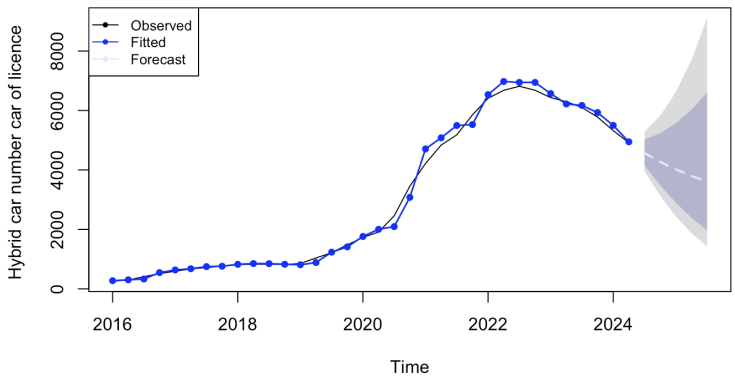

Figure 13: Forecast from damped Holt's method for hybrid vehicle licenses (original scale)

Figure 13 shows how the model adapts to the decline in hybrid vehicle registrations, providing a short-term forecast that continues the downward trend. As expected with the back-transformation, the confidence bands widen due to the increased variance.

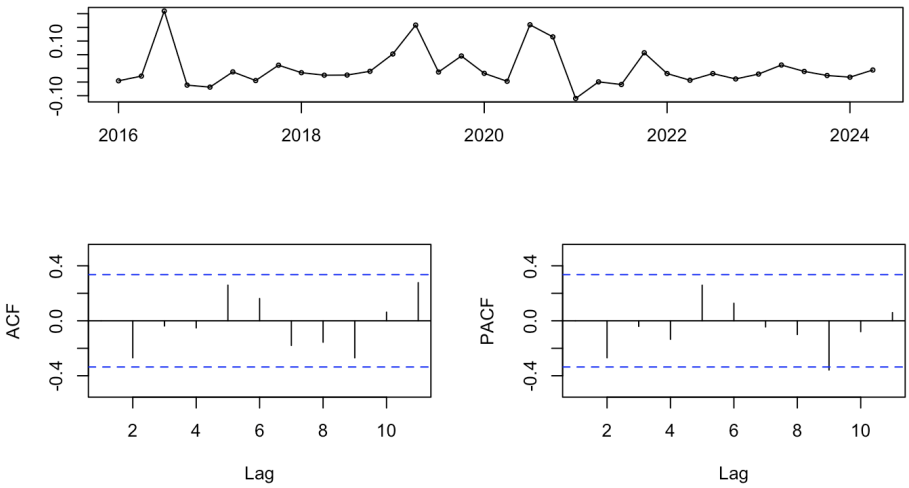

Figure 14: Residual diagnostics for damped Holt's method (hybrid series)

The residual analysis (Figure 14) shows no evidence of autocorrelation, in agreement with the p-value of the Ljung-Box test (0.3). The table below summarizes the forecast accuracy of exponential smoothing methods for hybrid series. The damped Holt model achieved the best values for RMSE,MAE and MAPE.

Table 2: Forecast accuracy metrics for exponential smoothing models – Hybrid series

Model	RMSE	MAE	MAPE	ACF1	Ljung-Box p	Theil's U
Holt Damped	179	125	4.82%	0.095	0.300	0.53
Holt	197	134	5.03%	0.097	0.335	0.55
Holt-Winters	227	149	5.66%	0.467	0.294	0.53

3.2.3 GAM

We started by transforming the series into logarithmic form to stabilize the variance. Then, we split the dataset into training and test sets to evaluate model performance. A Generalized Additive Model (GAM) was applied to the training data to capture nonlinear relationships.
The theoretical model is defined as:

$$\log(\text{hybrid}) = \beta_0 + f(\text{time}) + \varepsilon$$

The estimated model is the following.

$$\log(\text{hybrid}) = \beta_0 + s(\text{time}, k = 10) + \varepsilon$$

The smooth term for `time` is highly significant ($p < 2 \times 10^{-16}$), suggesting a strong nonlinear effect which explained most of the variability.

However, this model produced autocorrelated residuals, as indicated by the Ljung–Box test with its p-value(0.035). This implies that the residuals were not White Noise. To handle this, we modeled the residuals using an ARIMA(2,0,1)(1,0,0)[4] process, but the seasonal component did not show statistical significance. After this correction, the Ljung–Box test on the final residuals gave a p-value of 0.5806, indicating no remaining autocorrelation.

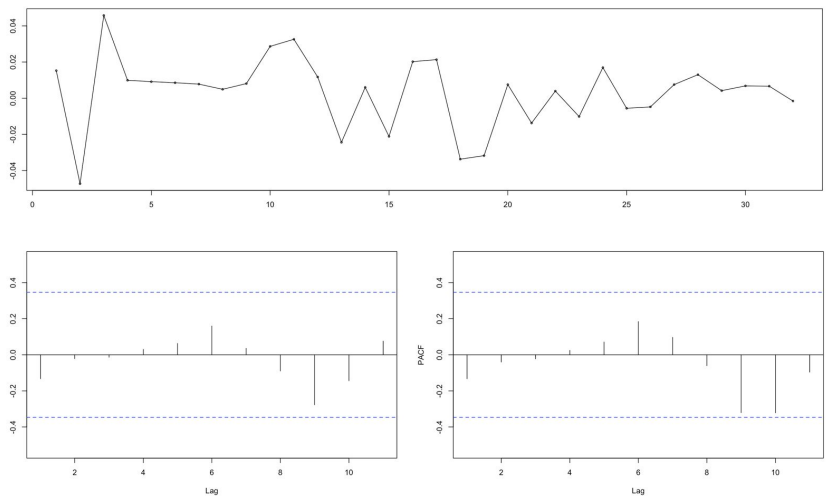

Figure 15: Residual diagnostics for ARIMA(2,0,1)(1,0,0)[4] model (hybrid vehicles)

The final residuals appear to be white noise, confirming the adequacy of the ARIMA correction.

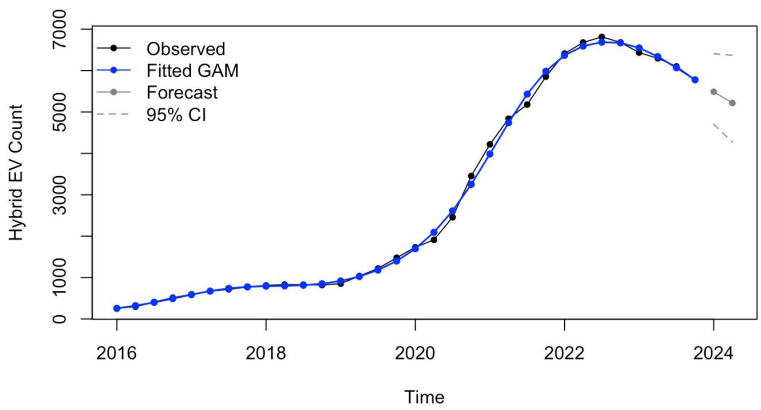

Figure 16: Forecast from GAM+ARIMA (residuals) method for electric vehicle licenses in thousands (original scale)

The plot shows the forecasted number of hybrid vehicle licenses using the GAM+ARIMA approach, which combines nonlinear trend estimation with residual temporal dynamics. The forecast is presented on the original scale, along with 95% confidence intervals derived from the combined uncertainty of both components of the model.

3.2.4 Arimax

The same approach used for the electric series was applied here. we started by searching for all relevant and available variables that could be useful for our forecasting problem. We created a function to detect and remove variables with high collinearity or multicollinearity by computing the VIF, excluding those with values greater than 5

Subsequently, with the remaining variables, we applied a backward stepwise selection to remove non-significant predictors from the model.
The theoretical ARIMAX model is defined as:

$$y_t = \beta_0 + \beta_1 \cdot \text{elec_consumption}_t + \beta_2 \cdot \text{charges}_t + \beta_3 \cdot \text{uk_petrol}_t + \beta_4 \cdot \text{cpi_energy}_t + \beta_5 \cdot \text{unemployment_uk}_t + \varepsilon_t$$

The estimated model is the following.

$$\hat{y}_t = 9.818 - 0.161 \cdot \text{elec_consumption}_t + 0.058 \cdot \text{charges}_t - 0.00110 \cdot \text{uk_petrol}_t + 0.026 \cdot \text{cpi_energy}_t + 0.418 \cdot \text{unemployment_uk}_t + \hat{\varepsilon}_t$$

Among the selected regressors, `charges` and `cpi_energy` are the strongest positive predictors of hybrid vehicle licenses adoption, both highly significant. `elec_consumption` and `uk_petrol` show significant negative effects, while `unemployment_uk` has a positive and significant association with the response.

In this case as well, the residual analysis revealed that the residual variance was not constant, indicating the presence of heteroscedasticity. If the primary objective were inference using multiple linear regression, we would consider applying a more accurate variance-stabilizing transformation. However, since our objective is forecasting, we proceeded by fitting an ARIMAX model to capture temporal dependencies and improve predictive accuracy. To further improve forecasting accuracy, we implemented a rolling window approach, which gave the best performance. We used a grid search to identify the optimal ARIMAX configuration. The selected model was ARIMA(1,1,0) process. The residuals from this model showed no significant autocorrelation or heteroscedasticity. The Ljung–Box test returned a p-value of 0.3097, confirming that the residuals resemble white noise and are well-behaved.

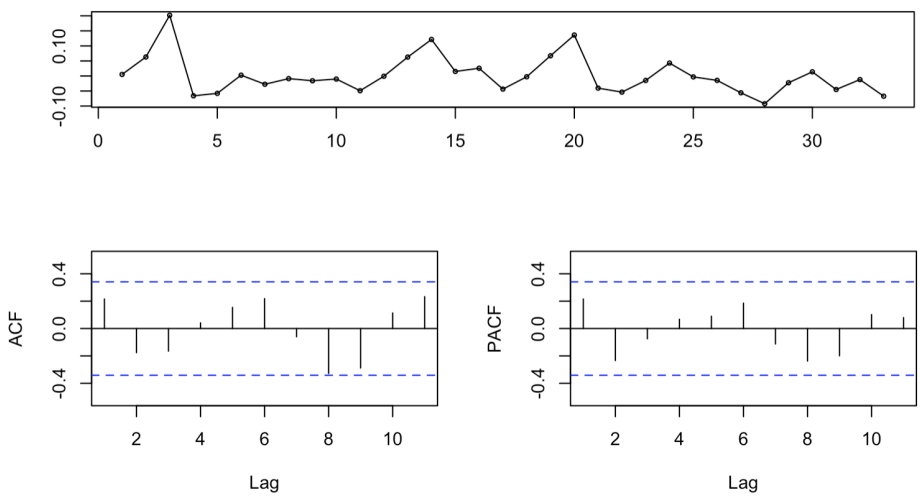

Figure 17: Residual diagnostics for ARIMA(1,1,0) (hybrid vehicles)

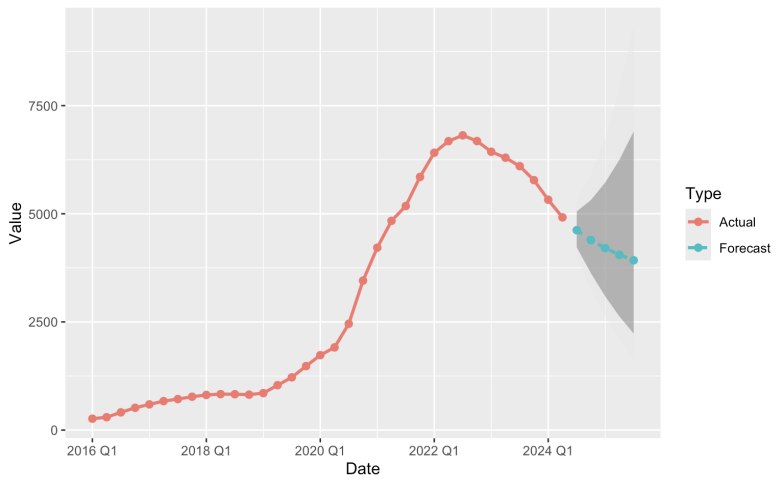

Figure 18: Forecast from ARIMAX model for hybrid vehicle licenses in thousands (original scale)

We can see a rapid increment of the hybrid vehicle licenses up to 2022, followed by a gradual decline. The ARIMAX model predicts a continuation of this downward trend, although the confidence intervals widen over time, reflecting increasing uncertainty in the forecast. Despite this, the model seems to adapt well to the recent dynamics and could serve as a reliable tool for short-term forecasting in this context.

4 Conclusion

This study compared various approaches to forecast the number of electric and hybrid vehicle licenses in the UK. Across both series, ARIMA-based and exponential smoothing models provided solid baseline performance, though they showed limitations in capturing nonlinear effects and the influence of external drivers, which were addressed by ARIMAX and GAM models.

In particular, ARIMAX models performed well when combined with a rolling window framework, effectively adapting to evolving dynamics. The GAM+ARIMA approach offered additional flexibility in modeling complex patterns, especially for the electric vehicle series.

Overall, the results underscore the importance of integrating statistical structure, external regressors, and nonlinear effects to enhance forecasting accuracy.

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Appendix A – Note on the Bass Model

While the number of electric and hybrid vehicle licenses might naturally suggest the use of diffusion models, such as the Bass model or its generalized versions, we chose not to include this approach in the main analysis. This decision is due to the fact that the available time series contain negative instantaneous values in some periods, which violate the non-negativity constraint.

For the interested reader, an exploratory implementation of the Bass model where negative values have been truncated to zero is provided in the accompanying code repository. This alternative modeling attempt is intended solely for illustrative purposes and is not used in the main forecasting evaluation, as it would be methodologically inappropriate.

Appendix B – Note on Data Aggregation

Some of the exogenous variables used in the forecasting models were originally available at a monthly frequency. To ensure consistency with the quarterly frequency of the response variable, these monthly series were aggregated to a quarterly level during the preprocessing phase. This transformation allows for coherent temporal alignment across all variables while preserving the essential dynamics of the original data.